

Logistic regression

The sigmoid of a linear score, its likelihood and gradient, softmax, logits, regularisation and imbalance.

THE QUESTION

Labels are 0 or 1. How do we turn a linear score into $P(y = 1 | x)$, and how do we fit it?

THE INTUITION

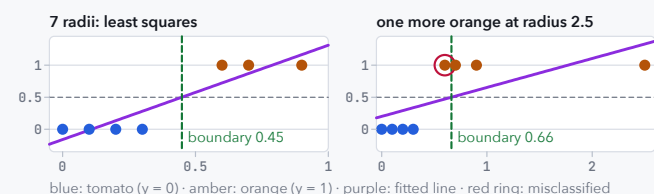
Pass the score $z = x^T \theta$ through the sigmoid into $(0, 1)$. Treat each label as a coin flip with that probability and pick the θ that makes the labels most likely: minimise the negative log-likelihood, the cross-entropy.

KEY TAKEAWAY

$P(y = 1 | x) = \sigma(x^T \theta)$: log-odds linear in x , boundary $x^T \theta = 0$. Minimise cross-entropy with gradient $X^T(\sigma(X\theta) - y)$; there is no closed form, so iterate.

score $z = x^T \theta \rightarrow \sigma(z) = P(y = 1 | x) \rightarrow$ Bernoulli likelihood $\rightarrow$ cross-entropy $J(\theta) \rightarrow$ step on $X^T(\sigma(X\theta) - y) \rightarrow$ boundary $x^T \theta = 0$

1 A straight line is a poor probability



The fruit notebook's seven radii, $y = 1$ (orange) above 0.4. Least squares gives $\hat{y} = -0.160 + 1.471x$, a boundary at 0.449 where $\hat{y} = 0.5$, and outputs outside $[0, 1]$. One more, obvious orange at radius 2.5 flattens the line: the boundary jumps to 0.663 and the orange at 0.6 becomes a tomato. The logistic boundary stays at 0.460.

2 The sigmoid makes the log-odds linear



$\sigma(z) = 1 / (1 + e^{-z})$ tends to 1 and 0 at the ends and $\sigma(0) = 0.5$. With $P(y = 1 | x) = \sigma(x^T \theta)$,

$$\text{odds} = \frac{P(y = 1)}{P(y = 0)} = e^{x^T \theta}, \quad \log \text{odds} = x^T \theta.$$

A unit more of x_j multiplies the odds by e^{θ_j} . $P(y = 1) = P(y = 0)$ where $x^T \theta = 0$: the boundary is a hyperplane.

3 Coin-flip likelihood gives cross-entropy

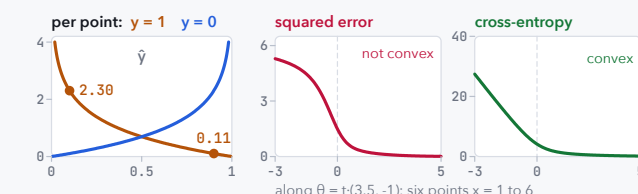


Ten flips with 4 heads have likelihood $\theta^4 (1 - \theta)^6$, largest at $\theta = 0.4$. Give point i its own coin, $\sigma(x_i^T \theta)$, and minimise the negative log-likelihood, the cross-entropy:

$$J(\theta) = - \sum_{i=1}^N [y_i \log \sigma(x_i^T \theta) + (1 - y_i) \log(1 - \sigma(x_i^T \theta))].$$

A Gaussian likelihood gave the SSE the same way.

4 Cross-entropy is convex, the SSE is not



For $y = 1$ a point costs $-\log \hat{y}$: 0.105 at $\hat{y} = 0.9$, 2.303 at 0.1, unbounded as $\hat{y} \rightarrow 0$. The squared error $\sum_i (y_i - \sigma(x_i^T \theta))^2$ on the cost notebook's six points is flat for bad θ , then bends down, so descent can stall. Cross-entropy is convex (the convexity and Newton's method sheet): any minimum is global.

5 The gradient is $X^T(\sigma(X\theta) - y)$

$X (7 \times 2)$		$\hat{y}$	y	$\hat{y} - y$
1	0	0.5	0	+0.5
1	0.1	0.5	0	+0.5
1	0.2	0.5	0	+0.5
1	0.3	0.5	0	+0.5
1	0.6	0.5	1	-0.5
1	0.7	0.5	1	-0.5
1	0.9	0.5	1	-0.5

grad = $X^T(\hat{y} - y)$
 θ_0 : sum of $(\hat{y} - y)$
 $= 4(0.5) - 3(0.5) = 0.5$
 θ_1 : sum of $x(\hat{y} - y)$
 $= 0.5(0.6) - 0.5(2.2) = -0.8$
 a step against it raises θ :
 bigger radius, higher $P(\text{orange})$

the 7 radii at $\theta = (0, 0)$: every $\hat{y} = \sigma(0) = 0.5$

Since $\sigma' = \sigma(1 - \sigma)$, the σ factors cancel in the chain rule:

$$\frac{\partial J}{\partial \theta_j} = \sum_{i=1}^N (\sigma(x_i^T \theta) - y_i) x_{ij}, \quad \nabla J = X^T(\hat{y} - y).$$

Least squares' $X^T(X\theta - y)$ with $\sigma(X\theta)$ for $X\theta$. The σ makes $\nabla J = 0$ nonlinear in θ : no closed form.

6 Gradient descent settles the boundary

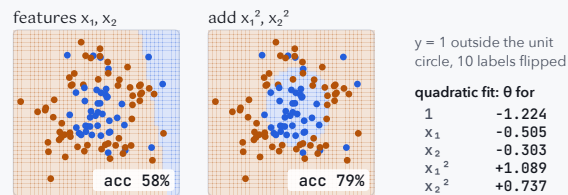


Our 60 points in two overlapping clouds (illustrative). Step $\theta \leftarrow \theta - \alpha/N \cdot X^T(\sigma(X\theta) - y)$ from zero: the mean loss falls from 0.693 to 0.27 in 20 steps and ends at 0.267 with $\theta = (-0.328, 2.18, 0.683)$. Newton's method needs a few steps (the next sheet, with IRLS).

Features, softmax and fixes for real data

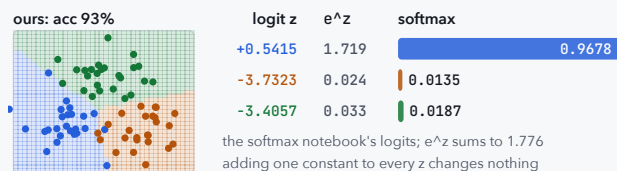
Curved boundaries, many classes, why losses take logits, separable classes and imbalanced classes.

7 New features bend the boundary



The circular notebook's 100 points. A line on (x_1, x_2) scores 58%, no better than always guessing the majority class: high bias. With x_1^2, x_2^2 added, $x^T \theta = 0$ is an ellipse: 79%, and the notebook's coefficients. Any basis from the linear regression II sheet works; the model stays linear in θ .

8 Softmax gives K probabilities that sum to 1

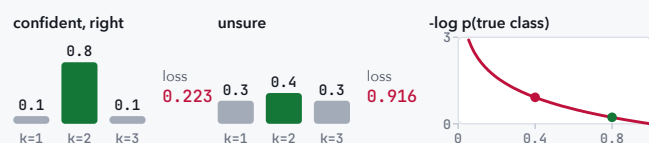


One column θ_k and one score $z_k = x^T \theta_k$ per class:

$$P(y = k | x) = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}}$$

The top score wins; neighbouring classes meet along lines. At $K = 2$ it is the sigmoid of $x^T(\theta_1 - \theta_0)$. Alternatives: one-vs-rest, one-vs-one.

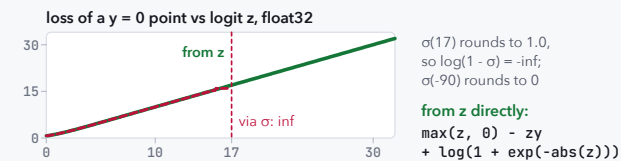
9 Cross-entropy for K classes



With one-hot y_i , $J = -\sum_i \sum_k y_{ik} \log \hat{y}_{ik}$, minus the log-probability of the true class: 0.223 and 0.916 for the slides' two predictions. The gradient is prediction minus target (the slides flip its sign):

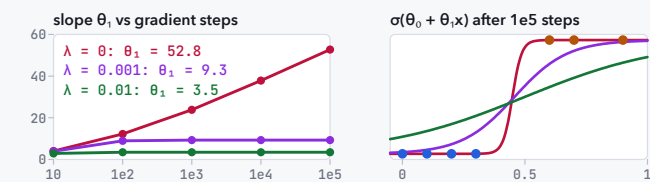
$$\nabla_{\theta_k} J = \sum_i x_i (P(y_i = k | x_i) - \mathbb{1}[y_i = k])$$

10 Losses take logits so nothing overflows



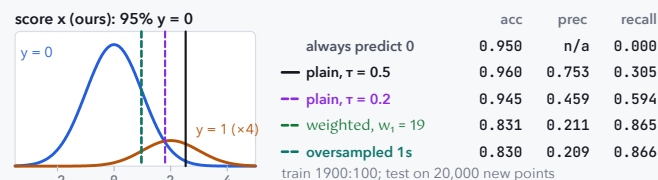
The raw score z is the logit. Both routes give 0.72693 on the logits notebook's example, but in float32 $\sigma(17)$ is exactly 1, so a $y = 0$ point costs $-\log 0$. From z the loss stays finite, so PyTorch's `binary_cross_entropy_with_logits` and `cross_entropy` take logits; softmax subtracts $\max_k z_k$ first.

11 Separable classes need regularisation



$J = J_1 + \lambda \theta^T \theta$ (L2) or $J_1 + \lambda \|\theta\|_1$ (L1), θ_0 unpenalised (the ridge and lasso sheets). If a line separates the classes, scaling θ up always lowers J_1 : no finite minimum. On the seven radii the slope climbs from 4.1 to 52.8 over 10 to 10^5 steps (the notebook's 56 is where its solver stopped); $\lambda = 0.001$ holds it at 9.3.

12 Accuracy hides a rare class



Always predicting 0 scores 95% and finds nothing; the plain fit gets 96% but recall 0.305. Each fix moves the cut left: a weighted loss ($w_i = N_0/N_1 = 19$ when $y_i = 1$), a threshold $\tau < 0.5$, or resampling (duplicate 1s, drop 0s, or add SMOTE's synthetic 1s). Judge by precision and recall; pick τ on held-out data (the precision-recall curves sheet).

13 Pause and think

Linear regression? Leaves $[0, 1]$; outliers move it.
The boundary? Where $x^T \theta = 0$.

Sigmoid or softmax? At $K = 2$ they agree.
Closed form? None; iterate.

Squared error? Not convex with σ .
What does λ buy? Finite θ when separable.

$P(y = 1 | x) = \sigma(x^T \theta)$ · minimise cross-entropy · $\nabla J = X^T(\sigma(X\theta) - y)$ · softmax for K classes · pass logits · regularise · reweight

Before: the lasso and coordinate descent sheet; next: the convexity and Newton's method sheet (Hessian $X^T S X$, Newton steps, IRLS). Source: lecture slides "Logistic regression" with the vectorisation notes, the logistic regression notebooks (fruit radii, circular data, cost surfaces, Iris, PyTorch) and the logits notebook (nipunbatra.github.io/ml-teaching). The 2D, three-class and imbalanced data are ours, seeded. Corrected: the coin-flip derivative and multi-class gradient signs, and the cost notebook's loss surfaces (a broadcasting slip).